

ICWaR Course

WR 202

Title: Geodetic Signal Processing (3:0)

Instructor: Dr. Bramha Dutt Vishwakarma

Term: January - May

Contents:

- A brief introduction to physical and satellite Geodesy, Geodetic data and Earth's surface processes,
- data from GRACE satellite mission, Spherical harmonic analysis and synthesis, GRACE data processing,
- Introduction to filtering, Kalman filter, Regression, time-series decomposition, moving window averages, introduction to data assimilation, Bayesian inference
- Global mass change trends, estimating Ice-sheet mass change, estimating groundwater change from satellites, closing the water and sea level budget.